Tai, Wei-Hsuan

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Education

National Taiwan University (NTU) Sep 2023 – Jun 2027 (expected)
B.S. in Biomedical Engineering GPA: 3.56/4.30 (last 3 semesters: 3.76/4.30)

- **Relevant coursework:** Machine Learning (A+), ML for Human Motion Analysis (A+), Deep Reinforcement Learning (A-), Data Structures (A+), Algorithms (A-), Programming Languages (A+), Database Management (A+), Fundamentals of Biomedical Image Processing (A), Principles of Medical Imaging Systems (A), Engineering Mathematics (A/A)
- **In progress (Fall 2026):** Multimodal Artificial Intelligence; Time-Frequency Analysis and Wavelet Transform
- Completed 25 credits of English-taught (EMI) courses

Publication

A multi-task masked autoencoder with GAN-based augmentation for PD-L1 prediction from chest CT images.
Scientific Reports, 2026. (4th author) doi:10.1038/s41598-026-49064-3

Research & Work Experience

Undergraduate Researcher, MIRAGE Lab, NTU (Advisor: Prof. Chung-Ming Chen) Jul 2025 – Present

- **PD-L1 prediction from chest CT (published above):** reproduced the proposed MAE+ViT pipeline with 9-fold fine-tuning; produced ROC/PR curves, AUC/AUPRC results and Wilcoxon signed-rank tests for the rebuttal.
- **Lesion CT texture synthesis (undergraduate thesis project):** explored improved DDPM-based methods for generating realistic lung-lesion textures on CT images.

Intern, Next-Gen Digital Healthcare Platform Project Office (MOHW) Jul 2026 – Present

- Researching and building an LLM-based clinical assistant that retrieves patient data from FHIR servers and supports differential diagnosis, missed-diagnosis prevention, examination recommendations, and draft medical-record generation.

Selected Projects

Deep Reinforcement Learning for Compiler Optimization Pass Ordering in LLVM 2026 Spring

- DRL course paper (OpenReview-style peer review + poster). Showed via ablation that pass-history state encoding beats IR-based features; built a multi-suite PPO pipeline (truncation-aware GAE, parallel rollouts) with an inference-time policy ensemble that matches or beats `-Oz` on 20/23 cBench programs.

Gait Biomechanics: Predicting COM-COP Displacement from Ground Reaction Forces 2026 Spring

- Compared BiLSTM, TCN and causal CNN with knowledge distillation under leave-one-subject-out CV; designed the preprocessing pipeline.

Lightweight Brain Tumor Segmentation via Knowledge Distillation 2025 Fall

- Distilled a trained nnU-Net into a lightweight 2.5D MobileNet-based U-Net on BraTS; distillation raised the student's mean Dice from 0.848 to 0.912 (5-fold CV, WT/TC/ET).
- https://bit.ly/4lwuwQy

Document Text Recovery with a Denoising U-Net 2025 Fall

- Replaced the assigned CLAHE baseline with a denoising U-Net trained on synthetically degraded images.
- https://github.com/W-X-Dai/text_recovery

Uniconnect – University–Industry Resource Platform (DBMS course) 2025 Fall

- Led backend: NestJS + PostgreSQL schema design, Redis caching, API integration, Docker deployment.
- https://github.com/W-X-Dai/DBMS_FinalProj

Leadership & Teaching

Vice President & Director of Events, NTU Biomedical Engineering Student Association [Term]

- Hosted department events including the graduation (tassel-turning) ceremony, freshman welcome party, and BBQ.

Instructor, NTU BME Camp for high-school students 2026 Summer

- Taught a 1-hour online lecture + Colab lab on deep learning for medical imaging (GAN augmentation, MAE, transfer learning).

Technical Skills

Programming: Python, C++ (Codeforces rating 1400), TypeScript/JavaScript, SQL
ML / DL: PyTorch, diffusion models, self-supervised learning, LLM agents (LangGraph), statistical evaluation
Backend / Tools: FastAPI, NestJS, PostgreSQL, Redis, FHIR, Vue, Linux, Docker, Git